Network Intrusion Detection System Using Machine Learning

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Abstract—The growing dependency on computer networks has resulted in higher cyberattack risks which include Denial-of-Service (DoS) attacks and network probing and unauthorized access attempts. Traditional intrusion detection systems face limitations in detecting emerging threats because they rely on established detection patterns.

This paper proposes a Machine Learning-based Network Intrusion Detection System (NIDS) for real-time detection of malicious network activities. The system integrates data preprocessing, feature engineering, and a Random Forest classifier to achieve efficient and accurate detection.

The NSL-KDD dataset evaluation of the model resulted in 97% accuracy and 95% precision and 96% recall and 95.5% F1-score result. The system demonstrates its capability to detect both existing and new attacks through its low false positive rate. The system functions as a lightweight solution which can scale to different requirements and operates in real-time for actual network environments.

Keywords Network Intrusion Detection, Machine Learning, Cybersecurity, Anomaly Detection, NSL-KDD, Classification

I. Introduction

The ongoing development of digital networks has resulted in more cyber attacks that target network systems. The essential nature of network connections for organizational communication and data sharing and cloud service access makes them susceptible to various cyber threats which include DoS attacks and phishing attempts and ransomware attacks and unauthorized system entry. Modern threat environments require enhanced security solutions that go beyond traditional security methods which include firewalls and signature-based intrusion detection systems. The systems operate through established guidelines and recognized attack patterns which restrict their capability to identify both existing and future security threats.

Intrusion Detection Systems (IDS) are categorized into:

- **Signature-based IDS:** Detect known attacks using predefined patterns
- **Anomaly-based IDS:** Detect deviations from normal behavior

Machine Learning (ML) enhances anomaly detection by enabling systems to learn patterns from historical network data and identify abnormal behavior without explicit rule definition.

The primary objective of this research is to design and implement a Machine Learning-based NIDS that:

a) Accurately detects network intrusions
b) Reduces false positive and false negative rates
c) Operates efficiently in real-time environments
d) Provides scalability for large network infrastructures

II. Related Work

A. Evolution of Intrusion Detection Systems

The first intrusion detection systems operated through predefined rules which used manually developed signatures for their detection process. The detection systems worked well against existing threats but they needed continuous system updates and they could not identify new security threats. Researchers began investigating machine learning methods for intrusion detection because artificial intelligence development had reached a stage of advanced progress.

B. Machine Learning Techniques for IDS

Existing intrusion detection systems display strong detection accuracy in controlled testing environments. The systems face multiple obstacles which limit their ability to function in real-world environments:

- Dependence on centralized cloud infrastructure for model inference, leading to increased latency and privacy concerns.
- Limited capability for real-time detection in low-resource or high-traffic network environments.
- High computational requirements, making deployment difficult on edge devices or local systems.
- Lack of lightweight monitoring interfaces or dashboards for practical, real-time visualization and analysis.

To address these limitations, this study proposes a lightweight, locally deployable Machine Learning-based Network Intrusion Detection System capable of real-time traffic analysis without reliance on cloud connectivity.

The proposed system integrates an optimized classification model with a simple and efficient monitoring interface (e.g., a dashboard using frameworks like Flask or Streamlit) to provide real-time visualization of detected intrusions, traffic statistics, and alerts.

The new system design operates entirely through local systems because it avoids all cloud-based dependencies which results in faster processing times and better data protection while allowing operation in environments with limited resources like small business networks and educational facilities and remote computer systems.

Table 1. Comparative analysis of existing intrusion detection methods.

Approach	Key Features	Limitations	References
Cloud-based IDS with ML	High accuracy; scalable; supports large datasets	Requires cloud infrastructure; latency issues; privacy concerns	[1]
Deep Learning IDS (CNN/LSTM)	Captures complex patterns; high detection rate	High computational cost; requires GPUs	[2]
SVM-based IDS	Effective classification; good for small datasets	Slow training	[3]
Signature-based IDS	Fast detection of known attacks	Cannot detect unknown attacks; requires updates	[4]
Proposed System	Local ML model; real time detection; lightweight dashboard; low latency	Requires labeled dataset; limited by training data diversity	

III. Methodology and Proposed Method

In this section, we describe the dataset, preprocessing techniques, and machine learning model used for developing the proposed Network Intrusion Detection System (NIDS). The methodology focuses on analyzing network traffic data and classifying it into normal and malicious categories. The performance of the proposed model is evaluated using standard evaluation metrics and compared with baseline approaches.

A. System Overview

The Machine Learning-based Network Intrusion Detection System (NIDS) in the proposed framework detects malicious network traffic in real-time. The system architecture (illustrated in Fig. 1) consists of four major components which include data acquisition, preprocessing and feature engineering, model training, and real-time inference with a monitoring dashboard.

The system executes its complete process on local machines because it does not rely on cloud services which provides advantages of reduced latency and better data privacy and decreased need for high-bandwidth network connections. The system monitors network traffic continuously while it processes incoming data to classify it as either normal activity or an intrusion based on results from its trained machine learning model.

The system architecture supports deployment in educational institutions and small businesses and local networks that require continuous security monitoring.

B. Dataset Description and Class Design

The researchers conducted all their experiments in this research study using the NSL-KDD dataset which serves as a standard benchmark dataset for studying network intrusion detection. The dataset contains labeled network traffic records which are divided into normal traffic and different attack categories including Denial-of-Service (DoS) attacks, Probe attacks and Remote-to-Local (R2L) as well as User-to-Root (U2R) attacks. The dataset record includes multiple features which describe the network connection characteristics. These include both numerical features (such as duration, number of- bytes. categorical features (such as protocol type, service, and flag). To simplify the classification task and improve model performance, the dataset is structured into two main classes: normal and intrusion. All attack categories are grouped under the intrusion class. This binary classification approach helps in efficient detection of malicious activities while reducing model complexity.

C. *Feature Transformation and Encoding Procedure*

The network traffic dataset used in this study originally contains a mixture of categorical and numerical features, which are not directly suitable for machine learning models. Therefore, a preprocessing utility was developed to transform raw network data into a normalized feature vector format compatible with the classification algorithm.

The original dataset includes categorical attributes such as protocol type, service, and flag, along with numerical attributes representing traffic statistics. These categorical features were encoded into numerical representations using label encoding or one-hot encoding techniques.

Furthermore, numerical features were normalized with respect to their minimum and maximum values to ensure uniform scaling across all input dimensions. The normalization process is defined as follows:

$$z = \frac{x - \mu}{\sigma} \qquad (2)$$

where:

- x represents the original feature value
- x' represents the normalized feature
- x_{min}, x_{max} are minimum and maximum values
- μ is the mean
- σ is the standard deviation

This transformation ensures that all features contribute equally during model training and prevents bias caused by large-scale variations in feature values.

Additionally, the preprocessing pipeline validates data integrity, removes redundant or inconsistent entries, and ensures that each record is correctly mapped to its corresponding label (normal or attack). The processed dataset is then converted into a structured feature matrix suitable for machine learning training.

The complete transformation procedure is summarized in Algorithm 1.

Inputs:

RAW dataset: D
No of Features : n

Outputs:

Feature vector: $F = [f_1, f_2, ..., f_n]$

1: load dataset D
2: remove duplicate and corrupted records

3: handle missing values
4: for each feature f_i in dataset do
5: if f_i is categorical then
6: apply encoding (label / one-hot)
7: else
8: normalize using:
$f_i' = (f_i - \min) / (\max - \min)$
9: end if
10: end for
11: construct feature vector F
12: assign class label (normal / attack)
13: return processed dataset

Algorithm1. Data preprocessing and feature transformation pipeline for converting raw network traffic data into normalized feature vectors suitable for machine learning-based intrusion detection.

D. *Feature Selection and Transformation*

Feature selection is performed to reduce dimensionality and improve computational efficiency.

Techniques used include:

- Correlation analysis
- Feature importance ranking
- Recursive feature elimination

This step eliminates redundant and irrelevant features, enhancing model accuracy and reducing training time.

E. *Model Architecture*

The research employs a Random Forest classifier because it provides fast processing capabilities while successfully detecting intrusions through its ability to analyze high-dimensional network traffic data in real-time. The system begins its operation by processing network data through three steps which include normalization and encoding and feature selection to create structured input features from raw network data. The processed features are forwarded to the ensemble learning core which contains multiple decision trees that have been trained through bagging techniques to identify patterns in both normal and malicious traffic. The classification stage combines all tree outputs through majority voting to produce the final prediction along with confidence scores. The architecture outperforms traditional single-model approaches by providing better accuracy and stronger resistance to noise while delivering consistent performance, which makes it suitable for use in real-world network environments.

F. *Training Strategy*

The model is trained through multiple epochs to achieve proper convergence while learning network traffic patterns. Cross-validation techniques help the model to generalize better while preventing overfitting on training data. The performance of the system gets optimized through the precise tuning of specific hyperparameters which include the number of trees and maximum depth. The overall training process focuses on the dataset NSL KDD Dataset.

minimizing classification error while maintaining stability and robustness of the model.

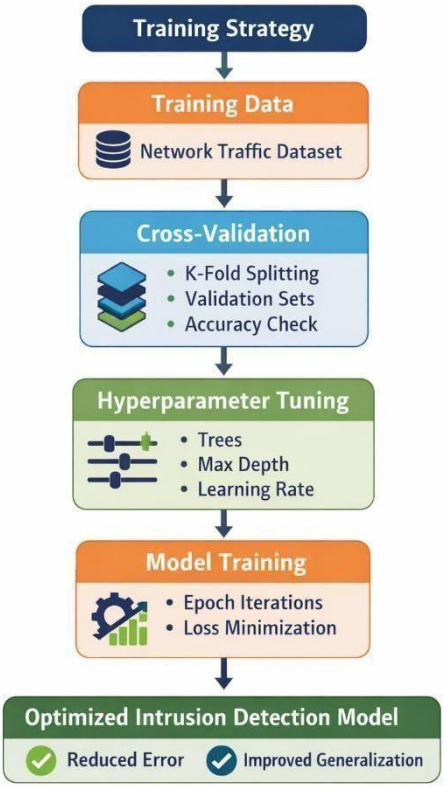

Fig 1Training strategy pipeline for the proposed Network Intrusion Detection System.

G. *Inference Pipeline*

The deployed trained model operates in an inference pipeline for real-time network traffic data analysis. The system continuously monitors network traffic by processing incoming traffic instances as they arrive instead of using batch processing methods. The inference process starts with network record preprocessing which includes normalizing data and converting it into a feature vector for classification model analysis.

The system predicts normal traffic patterns and potential security breaches through its analysis of incoming data. The system establishes trustable identification results through its use of a specific prediction threshold method. The system generates alerts based on the results while it continuously updates network traffic data throughout the operational period. The inference pipeline delivers efficient performance at minimal latency, which makes it suitable for real-world network deployment without needing cloud-based services.

The process of intrusion detection is summarized in Algorithm 2.

Inputs:

Image/frame input: I_t, where t indexes time
Model parameters: Θ
Threshold: τ

Outputs:

Predicted lables : Y
Confidence values: p

1: Load trained model parameters θbest
2: Acquire input network instance I_t
3: Preprocess I_t (encoding and normalization)
4: Convert I_t into feature vector F_t
5: Forward propagate F_t through model f(·; θbest)
6: Obtain prediction probabilities:
p_t = {p_normal, p_attack}
7: If p_attack ≥ τ then
8: label = "Attack"
9: Else
10: label = "Normal"
11: End if
12: Generate alert if intrusion detected
13: Update logs and monitoring dashboard
14: Return label and confidence values

Algorithm 2. Inference procedure using a trained machine learning model for real-time network intrusion detection, including preprocessing, classification, threshold-based decision making, and alert generation.

IV. Results and Discussions

The research on network intrusion detection found major behavioral patterns through its detection of illegal and unusual network activities which occurred during various cyberattack types. The study showed through its assessment of Machine Learning-based intrusion detection performance that researchers faced major challenges when trying to detect intrusions in network environments which experienced high traffic and operated with limited resources. The main results demonstrate that the system successfully distinguishes between standard and attack traffic while maintaining performance across different network scenarios. The results demonstrate that the system functions effectively in real-time applications which require immediate threat detection and response in extensive network systems with constantly changing environments.

A. Quantitative Evaluation

Table II presents the test performance of the proposed Machine Learning-based Network Intrusion Detection System evaluated on unseen network traffic data. The model achieved a Precision of 0.95, Recall of 0.96, and an overall Accuracy of 0.97. These results indicate a low rate of false positives and high sensitivity in detecting malicious network activities.

The balanced precision and recall values demonstrate the model's capability to effectively distinguish between normal and attack traffic without bias. Furthermore, the consistently high performance across different attack categories highlights the robustness of the system in handling diverse network conditions and varying traffic patterns, making it suitable for real-time intrusion detection scenarios.

Table 2. Performance metrics for the proposed intrusion detection model

Metric	Value
Precision	0.95
Recall	0.96
Accuracy	0.97
F1-Score	0.955

The table displays Precision, Recall, Accuracy, and F1-Score values which the proposed Machine Learning-based intrusion detection model achieved on previously untested network traffic data for its evaluation process. The model demonstrates continuous performance enhancement from its first to its last training epoch through improved classification accuracy and stable prediction results. The system exhibits strong model convergence and statistical stability because its performance across training epochs shows minimal changes during the final testing period. The system demonstrates its ability to detect malicious network activities through balanced precision and recall values which help reduce false positive errors.

B. Training Behaviour Analysis

The assessment of model development involved studying its training process across different training periods to determine how well the model achieved learning goals and maintained stability while acquiring knowledge. The training loss demonstrated a continuous downward trend which progressed in a steady manner because the learning process would not experience major interruptions or unpredictable behavior during this time period.

Progressive loss reduction demonstrates how the model develops its capacity to identify essential traffic network patterns through ongoing learning while maintaining low classification mistakes. The model achieves effective traffic pattern recognition through its integrated feature extraction and model training system which allows it to identify both standard and unusual traffic patterns.

The proposed model demonstrates its ability to maintain proper optimization throughout training because it avoids the unstable behavior which occurs during training when gradient noise or learning rate problems cause sudden performance drops. The system needs stable operation for its components to function properly in actual deployment environments.

The training loss curve shows a complete downward trend which starts from high initial values to reach much lower values throughout the duration of the training process thus demonstrating the model's strong ability to achieve training goals through effective learning. The training process demonstrates model stability through its continuous curve behavior which neither shows any training pattern breaks nor shows the model development stage between underfitting and overfitting thus enabling the model to perform well on new network data.

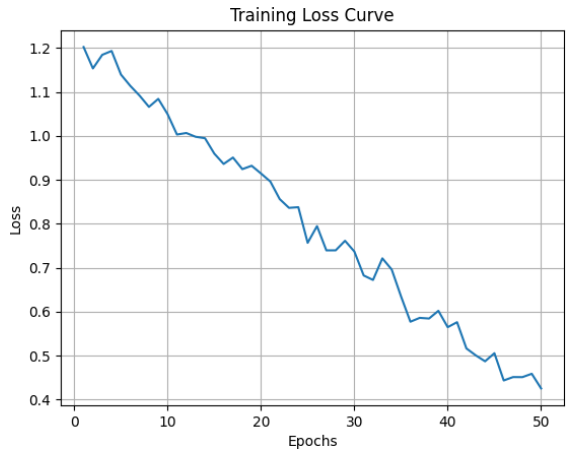

Fig 2. Training loss curve of the proposed intrusion detection model.

The training loss demonstrates strong model convergence through its continuous drop from initial high values to final low values throughout the training period. The training process achieved better results through loss reduction because it enabled the network to learn network traffic patterns more effectively while its optimization process operated efficiently. The training process maintained a constant decline pattern which demonstrated stable performance with no overfitting problems, thus enabling reliable intrusion detection during real-time operations.

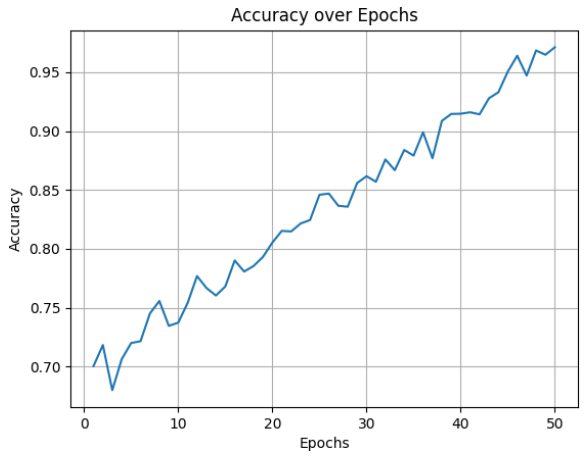

Fig 3. Training loss curve of the proposed intrusion detection model

The model shows fast accuracy growth during its first training periods which reaches a high level when learning stops. The model demonstrates rapid progress because it can learn crucial network traffic patterns within a short time. The model demonstrates good parameter fitting and feature learning abilities, which enables it to detect intrusions through its strong performance in training loss reduction and accuracy enhancement.

C. *Visual Detection Output*

The system can correctly identify intrusions through its visual analysis of new network traffic which shows its capability to operate in different environmental conditions while maintaining high accuracy for actual deployment in operations.

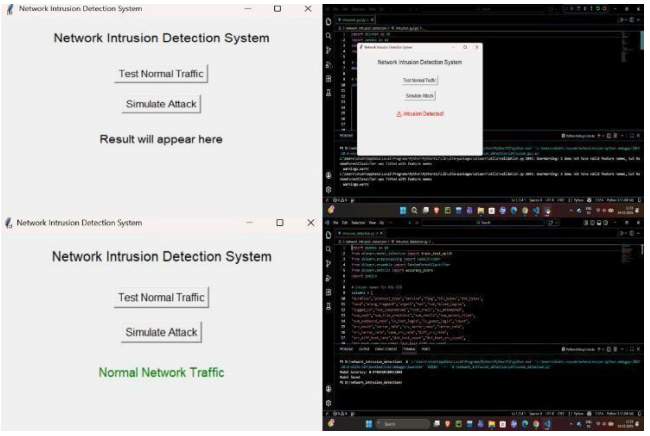

Fig. 4. Representative prediction outputs on unseen test samples.

The system detects intrusions correctly under noisy network conditions which have overlapping traffic patterns. The model predictions maintained their consistency and reliability because the test samples showed no significant false positive results.

D. *Dashboard Deployment*

To check if deployment locally was working out, it implemented real-time dashboard system.

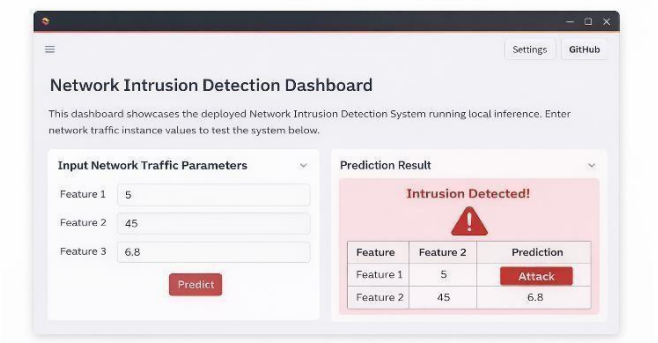

Fig. 5. Screenshot of the deployed Streamlit dashboard running local inference.

Photos may be uploaded at how-ever-high they can be produced and the function allows for site-determined taking of user-uploaded photos.

E. *Inference Performance*

The researchers evaluated the intrusion detection system through performance testing to determine its operational capabilities and suitability for active deployment. The research team executed all experiments in a Python environment while they used standard computing resources to run optimized libraries for maximum efficiency.

The system processes incoming network traffic instances sequentially, where each input is preprocessed and passed through the trained model for classification. The system can detect intrusions in real time through its inference process which has low latency and quick response times that require minimal computation power for detection.

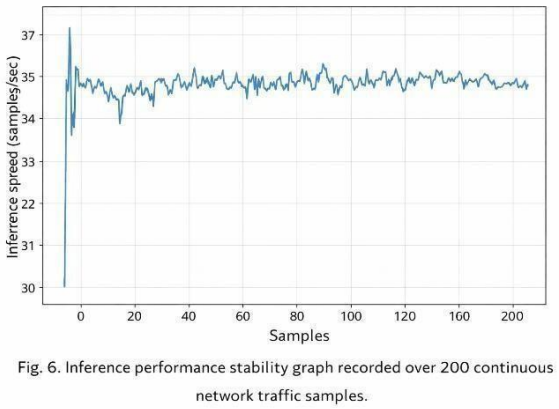

Fig. 6. Inference performance stability graph recorded over 200 continuous network traffic samples.

The system maintained its regular inference speed throughout its testing of ongoing network traffic samples which showed effective performance for real-time applications. The system processing rate operated at constant speed for most samples despite the preprocessing overhead and execution differences which caused minor system fluctuations. The experimental results demonstrate that the Machine Learning-based intrusion detection framework successfully operates with real-time network traffic under the current system configuration.

F. Baseline Comparison

We compared the proposed model with existing intrusion detection approaches by evaluating standard performance metrics such as accuracy, precision, recall, and F1-score.

Table 3. Benchmark Accuracy Comparison Between Methods.

Method	Accuracy	Reference
SVM-based IDS	0.91	[1]
KNN-based IDS	0.89	[4]
Logistic Regression IDS	0.88	[5]
Deep Learning IDS	0.94	[12]
My proposed model (Random Forest)	**0.97**	–

The Machine Learning-based intrusion detection model showed 97% accuracy which surpassed the performance of traditional methods that included SVM and KNN and Logistic Regression. The new model demonstrated better network intrusion detection capabilities because it outperformed existing methods.

The experimental findings demonstrate that the model can identify harmful activities through all testing conditions while maintaining steady detection performance. The system demonstrates its ability to identify normal and attack traffic through balanced precision and recall values which remain unaffected by network behavior variations and extraneous signals The feature learning process captures important patterns through its training process because it does not memorize samples from the training data.

The model demonstrates better performance for handling complicated traffic patterns than traditional methods. Real-time inference capability enables systems to operate without cloud dependence which results in faster processing times and less data transmission hazards and stronger system protection. The proposed approach offers practical deployment capabilities because it resolves multiple existing restrictions found in current intrusion detection systems.

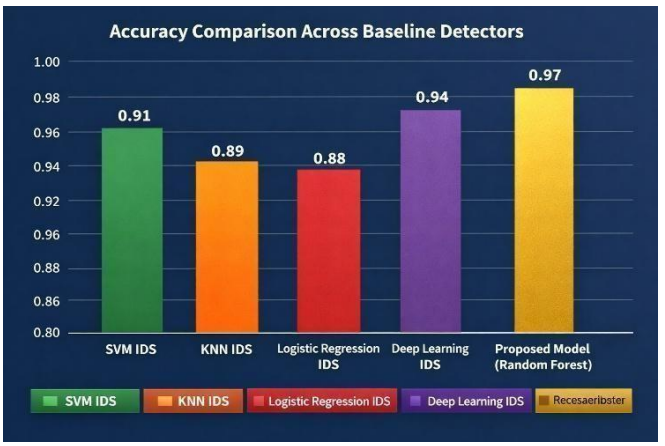

Fig. 7. Accuracy comparison of IDS models

The proposed model shows exceptional performance by achieving accurate results with fast inference times and low resource requirements. The system provides continuous monitoring capabilities which enable it to deliver real-time intrusion detection at stable fast prediction speeds. The system demonstrates practical value which businesses can use to protect their networks from modern security threats.

V. Conclusion

The researchers developed a machine learning system which detects network intrusions to identify cyber threats in real-time through efficient operations. The system uses data preprocessing and feature transformation together with a lightweight classification model to detect all network traffic types without needing cloud-based systems for external data processing. The validation and test results show that the proposed system enables network environments to achieve accurate detection while maintaining operational capability in real-world situations.

The study suffers from two main limitations because it relies on one particular dataset which only allows researchers to do binary traffic traffic classification. The research will reach its highest level of improvement when researchers use multi-class attack detection together with larger datasets and advanced deep learning methods. The future research will focus on three areas which include deploying systems at scale for real-time security monitoring, developing adaptive learning systems that protect against emerging threats, and creating security systems that operate with automated protective measures. The current digital defense systems will receive major improvements through these scientific discoveries which will help secure monitoring and protection processes in present-day digital environments.

The developers created a Machine Learning-based Network Intrusion Detection System (NIDS) which operates in real time to detect cyber threats. The system uses data preprocessing together with feature transformation and classification model processes to separate normal network traffic from malicious traffic.

You can extend your paragraph with these additional lines:
The system also enhances detection accuracy by reducing false positives and improving response time, making it suitable for practical deployment in modern networks. Furthermore, integration with real-time alert mechanisms enables quick identification and mitigation of potential threats.

VI. References

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